Name:_____ Period:_____
Earth and Space Science

ENERGY IN EARTH'S SYSTEMS AND HEAT TRANSFER LEARNING GUIDE

Learning Intention
Today we will be learning how energy from the Sun and from Earth's interior moves through Earth's systems by conduction, convection, and radiation so we can explain what powers Yellowstone's hot springs, the winds over Yellowstone Lake, the slow movement of Earth's mantle, and why melting ice can lead to even more warming.

Success Criteria
I can… ● Identify Earth's two energy sources and explain the difference between heat and temperature. ● Classify examples of conduction, convection, and radiation and use density to explain a convection current. ● Use data to explain how specific heat and changes in reflection, like melting ice, affect how Earth's surface heats up.

Yellowstone: A Park Full of Energy

Yellowstone National Park, located mostly in Wyoming, has more geysers and hot springs than anywhere else on Earth. Old Faithful erupts about every 60 to 110 minutes. In winter, snow covers the park and Yellowstone Lake freezes over, but steam still rises from the hot springs. Few places on Earth show energy moving as clearly as Yellowstone does. Heat from deep inside Earth powers its geysers, while sunlight melts its snow and warms its lake. In this guide, Yellowstone is our example: every idea you learn can be seen somewhere in the park. As you read, keep this question in mind: *Where does all of this energy come from, and how does it move?*

Earth's Two Energy Sources

Energy is the ability to do work or cause change. Energy is never created or destroyed. It is transferred from place to place or changed from one form to another. Earth's systems get energy from two main sources:

- **The Sun (external source):** Energy from the Sun reaches Earth as electromagnetic energy. It heats Earth's surface, drives weather and ocean currents, and powers the water cycle. At Yellowstone, sunlight melts the snow in spring and warms the land and lake in summer.
- **Earth's interior (internal source):** Heat inside Earth comes mainly from the **radioactive decay** of unstable elements, with some heat left over from Earth's formation. This heat drives the slow movement of the mantle, which moves tectonic plates and causes volcanoes and earthquakes. At Yellowstone, a large body of hot, partly melted rock a few miles below the surface heats underground water and powers the geysers and hot springs.

Check for Understanding
1. Write **S** if the event is powered by the Sun or **I** if it is powered by Earth's interior. _____ a. Old Faithful erupts. _____ b. Snow melts in a Yellowstone meadow in May. _____ c. A hot spring stays hot all winter.

_____ 2. Heat inside Earth comes mainly from
(1) sunlight absorbed by rocks at the surface
(2) the radioactive decay of unstable elements
(3) friction from winds blowing across the land
(4) infrared energy reflected by clouds

Heat and Temperature

All matter is made of particles that are always moving. The energy of motion is called **kinetic energy**.

- **Temperature** is a measure of the average kinetic energy of the particles in a substance. When particles move faster, the temperature is higher.
- **Heat** is the transfer of thermal energy from one place to another. Heat always flows from an area of higher temperature to an area of lower temperature. It keeps flowing until both areas reach the same temperature.

Check for Understanding

_____ 1. Hot water from Yellowstone's geyser basins flows into the cooler Firehole River. Which statement describes the flow of heat?
(1) Heat flows from the river into the hot water.
(2) Heat flows from the hot water into the river.
(3) Heat flows equally in both directions.
(4) No heat flows because both are water.

Three Methods of Heat Transfer

- **Conduction** is the transfer of heat through direct contact, from particle to particle. It works best in solids, especially metals.
- **Convection** is the transfer of heat by the movement of a **fluid** (a liquid or a gas). Warm fluid rises and cool fluid sinks, forming a circular pattern called a **convection current**.
- **Radiation** is the transfer of energy by electromagnetic waves. Radiation does not need matter to travel, so it can move through the empty space between the Sun and Earth. Conduction and convection both need matter, so they cannot carry energy across space.

Method	How energy moves	Needs matter?	Yellowstone example
Conduction	Direct contact, particle to particle	Yes (works best in solids)	Hot ground warms the wooden boardwalk resting on it
Convection	Movement of a fluid due to density differences	Yes (liquids and gases)	Hot water rising in a hot spring
Radiation	Electromagnetic waves	No	Sunlight warming the snow

Check for Understanding

1. **Classify It.** Write **CD** (conduction), **CV** (convection), or **R** (radiation) for each example.

_____ a. Sunlight warms a hiker's face on a cold, clear morning.

_____ b. Air warmed above a hot spring rises and carries steam upward.

_____ c. A hiker's hands warm up while holding a mug of hot cocoa.

2. Explain why radiation is the only method that can transfer energy from the Sun to Earth.

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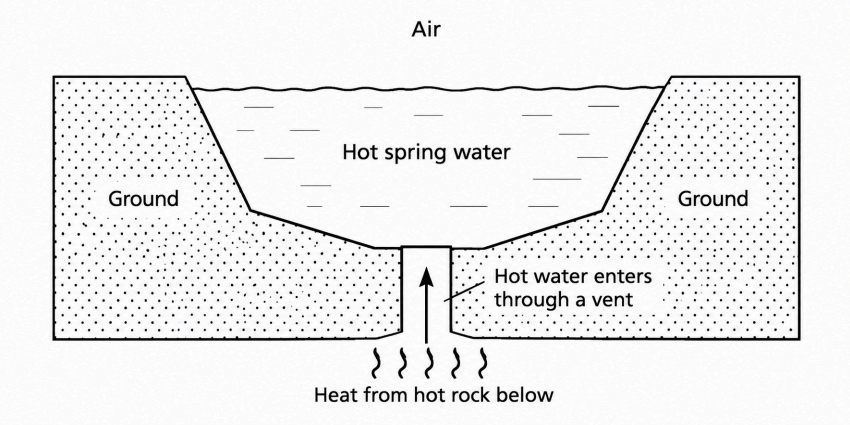
Convection in Earth's Systems

Convection happens anywhere material can flow. When a fluid is heated, it expands, becomes **less dense**, and rises. When it cools, it becomes **more dense** and sinks.

- **Atmosphere:** Air warmed by the ground rises and cooler air sinks, creating winds, clouds, and thunderstorms.
- **Oceans:** Differences in temperature and salinity make dense water sink and less dense water rise, moving heat around the planet.
- **Mantle:** Mantle rock is solid, but it is hot enough to flow very slowly over millions of years. Rock heated deep in the mantle becomes less dense and rises. Near the top, it cools, becomes more dense, and sinks. The part of the upper mantle where rock flows most easily, just below the lithosphere, is the **asthenosphere**. Mantle convection helps move tectonic plates. Many scientists think hot rock rising from deep in the mantle supplies the heat beneath Yellowstone.

Check for Understanding: Model It

The diagram represents a cross section of a Yellowstone hot spring. The water is heated from below and cools where it meets the cold air at the surface.



1. **Draw at least four arrows** on the diagram to show the convection current in the hot spring water. Then write **less dense** where the water is rising and **more dense** where the water is sinking.

2. Explain why the water near the surface at the edges of the spring sinks. Use the word *density*.

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_____

Energy from the Sun: Absorption and Reflection

Most of the Sun's energy reaches Earth as visible light. Some of it is reflected back to space and some is absorbed. Dark, rough surfaces, like Yellowstone's pine forests and bare soil, **absorb** more energy. Light, smooth surfaces, like fresh snow and the ice on Yellowstone Lake, **reflect** more energy.

After Earth's surface absorbs sunlight and warms up, it gives off energy as **infrared** radiation, which has a longer wavelength than visible light. Objects give off infrared energy all the time, even at night, and warmer objects give off more infrared energy than cooler objects.

Greenhouse gases in the atmosphere, such as carbon dioxide and water vapor, absorb some of this infrared energy and keep the atmosphere warmer. Scientists fly cameras that detect infrared energy over Yellowstone to map its hottest ground.

Check for Understanding

_____ 1. Which Yellowstone surface would most likely absorb the most sunlight?
(1) fresh snow in a meadow
(2) ice on Yellowstone Lake
(3) a dark pine forest
(4) light-colored hot spring deposits

2. Explain why an infrared camera can locate hot ground at Yellowstone even at night.

_____

_____

Specific Heat: Why Land and Water Heat Differently

Specific heat is the amount of energy needed to raise the temperature of 1 gram of a substance by 1°C. A substance with a **high** specific heat heats up slowly and cools down slowly. A substance with a **low** specific heat heats up quickly and cools down quickly.

Liquid water has one of the highest specific heats of any common substance, so land heats up and cools down faster than water. Yellowstone Lake is one of the largest high-elevation lakes in North America. It warms so slowly that the water stays cold all summer, even on days when the sandy shore feels warm. Large bodies of water keep the temperatures near their shores more moderate.

Check for Understanding

1. According to the reading, how does a substance with a low specific heat respond when energy is added?

_____

_____ 2. Compared to the land around it, Yellowstone Lake
(1) heats up faster and cools down faster
(2) heats up faster and cools down slower
(3) heats up slower and cools down faster
(4) heats up slower and cools down slower

Energy In, Energy Out, and Feedbacks

Earth's temperature depends on a balance between the energy coming in and the energy going out. Energy comes in from the Sun. Earth's surface and atmosphere send energy back out to space as infrared radiation. When the energy coming in equals the energy going out, Earth's average temperature stays about the same. If more energy comes in than goes out, Earth warms. If more energy goes out than comes in, Earth cools. For example, a large volcanic eruption can put ash and gases high in the atmosphere that reflect sunlight, cooling Earth for a year or two. Adding more greenhouse gases does the opposite: less infrared energy escapes to space, so Earth warms.

A **feedback** happens when a change causes effects that make the original change bigger or smaller. Melting ice is a feedback that makes warming bigger. Snow and ice reflect most of the sunlight that reaches them. When they melt, they uncover darker land or ocean water, which absorbs more sunlight. The surface warms even more, which melts even more ice. The same thing happens on a small scale in a Yellowstone meadow each spring: once patches of dark ground appear, the snow around them melts faster.

Check for Understanding

1. **Model It.** Complete the feedback loop for melting ice by filling in the two blanks.

Earth's surface warms → Snow and ice melt → _____

→ _____ → Even more snow and ice melt

Base your answers to questions 2 and 3 on the data table below, which shows the approximate average extent (area) of Arctic sea ice each September.

Year	1980	1990	2000	2010	2020
Sea ice extent (million km²)	7.7	6.1	6.3	4.9	3.9

Source: National Snow and Ice Data Center, rounded

_____ 2. Which statement best describes the change in September Arctic sea ice from 1980 to 2020?
(1) It increased steadily every decade.
(2) It stayed about the same.
(3) It decreased overall by about half.
(4) It decreased only between 1980 and 1990.

3. Use the data and the idea of feedback to explain how losing Arctic sea ice can cause even more warming.

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ESSRT Connection

Use the **Generalized Portion of Electromagnetic Spectrum Related to Earth and Space Sciences** (page 3) and the **Generalized Model of Earth's Interior Structure** and **Generalized Cross Section Model of Earth's Surface and Interior** (page 11) in the 2026 Revised Edition Reference Tables for Earth and Space Sciences.

1. The cameras that map Yellowstone's hot ground detect energy with wavelengths just longer than visible light. Use page 3 to identify this type of electromagnetic energy.

_____

_____ 2. Compared to infrared radiation, ultraviolet radiation has a
(1) longer wavelength and more energy
(2) longer wavelength and less energy
(3) shorter wavelength and more energy
(4) shorter wavelength and less energy

3. According to the Generalized Model of Earth's Interior Structure, what is the density range of the mantle?

_____ g/cm³

4. On the Generalized Cross Section Model, identify the layer directly below the lithosphere where mantle rock flows slowly in convection currents.

_____

Cluster Set 1

Base your answers to questions 1 through 4 on the information and data table below and on your knowledge of Earth and Space Science.

A student wanted to model why the sandy shore of Yellowstone Lake can feel warm on a summer afternoon while the lake water stays cold. The student placed equal masses of dry sand and water under the same heat lamp. The lamp was on for 12 minutes and then turned off. The student recorded the temperature of each material every 3 minutes.

Time (min)	0	3	6	9	12	15	18	21	24
Lamp	ON	ON	ON	ON	OFF	OFF	OFF	OFF	OFF
Sand (°C)	20	26	31	35	38	33	29	26	24
Water (°C)	20	21	23	24	25	25	24	24	23

_____ 1. Which statement best describes the data?
(1) The sand heated up and cooled down faster than the water.
(2) The water heated up and cooled down faster than the sand.
(3) The sand heated up faster, but the water cooled down faster.
(4) Both materials heated up and cooled down at the same rate.

2. Calculate the change in temperature of the sand and of the water from 0 to 12 minutes.

Sand: _____ °C Water: _____ °C

_____ 3. Which claim and supporting evidence best explain the difference in how the sand and water heated?

	Claim	Evidence
(1)	Water has a higher specific heat than sand.	With the same lamp and time, the water's temperature rose 5°C while the sand's rose 18°C.
(2)	Sand has a higher specific heat than water.	The sand reached a higher temperature than the water after 12 minutes.
(3)	The water received less energy than the sand.	The water started at the same temperature as the sand.
(4)	Water reflects all energy from the lamp.	The water's temperature did not change during the experiment.

4. Use the data to explain why the sand on the shore of Yellowstone Lake can feel warm on a sunny afternoon while the lake water stays cold.

_____

_____

Mini Glossary

Absorption – when a material takes in energy instead of reflecting it

Asthenosphere – the layer of the upper mantle, below the lithosphere, where rock flows slowly

Conduction – the transfer of heat through direct contact, from particle to particle

Convection – the transfer of heat by the movement of a fluid caused by differences in density

Convection current – the circular movement of a fluid as warm, less dense material rises and cool, more dense material sinks

Density – the amount of mass in a given volume of a material

Electromagnetic energy – energy that travels as waves and does not need matter to move

Energy – the ability to do work or cause change

Energy balance – the balance between the energy Earth takes in from the Sun and the energy it sends back to space

Feedback – a change that causes effects that make the original change bigger or smaller

Fluid – any material that can flow, such as a liquid or a gas

Greenhouse gas – a gas in the atmosphere, such as carbon dioxide or water vapor, that absorbs infrared energy

Heat – the transfer of thermal energy from an area of higher temperature to an area of lower temperature

Infrared – electromagnetic energy with a wavelength longer than visible light, given off by Earth's surface

Kinetic energy – the energy of motion

Radiation – the transfer of energy by electromagnetic waves

Radioactive decay – the breakdown of unstable elements that releases heat inside Earth

Reflection – when energy bounces off a surface instead of being absorbed

Specific heat – the amount of energy needed to raise the temperature of 1 gram of a substance by 1°C

Temperature – a measure of the average kinetic energy of the particles in a substance

Wavelength – the distance from one wave crest to the next